

ASSIGNMENT II

This assignment uses knowledge and skills from the previous few courses on Business Intelligence Solutions, Implementing Data Models and Reports, and Analyzing Data with Power BI. It is a continuation of the Dashboard created in Assignment I.

This assignment will serve as a practical and hands on application with real world data. This is an individual assignment. You may not work with other students.

For this assignment, you have already been provided with specific data by your instructor. You will use the data and the dashboard to create Data Models and Reports.

The assignment will be in two parts:
Assignment II Power BI to analyze and visualize data.
Assignment II Oral Presentation on Day 8

Your Dashboard will include the following:

In this assignment, you will continue with Power BI to analyze and visualize sales data from the "Global Superstore" dataset, which contains information about orders, customers, products, and sales. You will build a Power BI report and answer a series of questions to gain insights into the sales performance of the global superstore.

1. Total Sales Overview:

- Create a visualization that displays the total sales for the entire dataset.
- Calculate and display the total profit.
- Show the number of unique customers.

2. Sales by Region:

- Create a map visualization that displays total sales by region.
- Include data labels for each region.
- Add a slicer that allows users to filter by market.

3. Category-wise Analysis:

- Create a bar chart that shows total sales for each product category.
- Create a line chart showing the trend of profit over time.
- Add a filter for selecting specific categories.

4. Customer Segmentation:

- Create a pie chart that shows the distribution of customers by segment.
- Display the total profit generated by each customer segment.
- Add a slicer to filter by market.

5. Shipping Analysis:

- Create a scatter plot that visualizes the relationship between shipping cost and sales.
- Use color-coding to represent different ship modes.
- Add a trendline to identify any correlation.

6. Discount Impact:

- Create a matrix visualization to analyze the impact of discounts on profit.
- Calculate the average profit and sales for each discount range (e.g., 0-10%, 11-20%, etc.).
- Add a filter for selecting a specific category.

7. Customer Cohort Analysis:

- Build a cohort analysis that groups customers by their first purchase month and year.
- Calculate the average sales for each cohort over time.
- Highlight any insights about customer retention and growth.

8. Forecasting:

- Implement a forecasting model for future sales.
- Use historical sales data to predict sales for the next 12 months.
- Include a line chart to visualize actual vs. forecasted sales.

The reports will be prepared independently by each student, with guidance in the form of 1 private consultation with the instructor on either Day 6 or Day 7.

Your Oral Presentation will include the following:

Your models and reports will be presented orally to the class and instructor at an assigned time for 30 minutes, plus 15 minutes of questions.

Consultation: The student will meet with the instructor once, on Day 6 or Day 7, at a mutually agreed upon time, for 30 minutes, to receive tips, suggestions, and consultation on the assignment progress.

Assignment II Presentation

- 30 minutes on Day 8, using any multimedia or presentation tools or the student's choosing
- 15 minutes of questions from the audience
 - **The presenter must prepare questions to engage the class, in case no questions are asked**
 - Students not conforming to time requirements will be penalized by up to 10% depending on the scale of non-conformance, either too long or too brief

Assignment II Models and Reports Submission

- Models and Reports are due immediately following the oral presentation
- Must contain quality and appropriateness of visualizations, relevance of analysis, clarity and organization of the report.
- Must be not less than 500-750 words, 2 to 3 pages in length

- Deductions made if it is too long or too short
- Submit electronically
 - Students not conforming to these requirements will be penalized by up to 10% depending on the scale of non-conformance.

Marking Scheme

- 10% Consultation II with the instructor
- 50% Models and Reports submission
- 40% Oral Presentation of the Models and Reports, showing grasp and professionalism.
 - Marks are based on
 - Presentation skills
 - Quality of the presentation
 - Ability to engage the class with asking and answering questions
 - Demonstrate capability to present the Models and Reports with passion and knowledge to engage the audience
 - Communicate with ease and effective skills in order to reach the audience
 - Make the presentation both interesting and fun

Spelling and Grammar: Proofread your submission for spelling and grammatical errors.